CLEARED GUIDANCE AND QUANTUM-POTENTIAL IDENTITIES FOR THE POLAR FORM OF A WAVE FUNCTION OVER THE INTEGERS: A LEAN 4 FORMALIZATION

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Abstract. We record and machine-check two polynomial identities that underlie the de Broglie–Bohm guidance law and the Bohm quantum-potential prefactor, in a form cleared of all denominators. Writing $\psi = (Rc, Rs)$ and the product-rule derivative $d\psi = (dR\,c - Rs\,dS,\ dR\,s + Rc\,dS)$, we prove that $\mathrm{Im}(d\psi \cdot \overline{\psi}) = dS\,|\psi|^2$ holds as an identity in the integers $\mathbb{Z}$ for arbitrary values of R, dR, c, s, dS, and that $(-\Delta R)(2R) = 2\,((-\Delta R)R)$. The proofs are carried out in Lean 4 using the core library only (no Mathlib), contain no `sorry`, and depend on no axioms beyond Lean's standard ones. The paper states precisely what is formalized and what is not.

1. Introduction

In the Madelung (polar) representation $\psi = Re^{iS}$ of a wave function, the de Broglie–Bohm velocity field is $\nabla S = \mathrm{Im}(\nabla\psi/\psi)$ wherever $\psi \neq 0$, and the quantum potential is $Q = -\frac{\hbar^2}{2m}\,\Delta R/R$. Both statements are, at bottom, algebraic identities once the derivatives are expressed through the product rule. This note isolates those identities in their *cleared* (division-free) form and verifies them in the Lean 4 proof assistant.

The formal development is deliberately elementary: all quantities are integers and the identities are polynomial. The contribution is a short, fully checked statement of exactly which algebra the guidance law rests on. Analytic content (differentiability, the Schrödinger equation, winding numbers, the harmonic-oscillator spectrum, split-step integrators) is *not* formalized here.

2. Setting

Fix integers R, dR, c, s, dS. Here c, s play the role of $\cos S, \sin S$ (no relation $c^2 + s^2 = 1$ is assumed), dR and dS the derivatives of amplitude and phase along one direction. Put

$$\psi = (Rc,\ Rs), \qquad d\psi = (dR\,c - Rs\,dS,\quad dR\,s + Rc\,dS).$$

The second formula is the product rule for $R\cdot(c, s)$ with $dc = -s\,dS$, $ds = c\,dS$; in the formalization it is a *definition of the input*, i.e. a hypothesis on the shape of $d\psi$, not something derived.

For $z = (a, b)$, $w = (a', b')$ regarded as complex numbers $a+ib$, $a'+ib'$, one has $\mathrm{Im}(z\overline{w}) = ba' - ab'$ and $|w|^2 = a'^2 + b'^2$.

3. Results

Theorem 3.1 (Cleared guidance law; `Bohm.guidance_cleared`)**.** *For all $R, dR, c, s, dS \in \mathbb{Z}$,*

$$(dR\,s + Rc\,dS)(Rc) - (dR\,c - Rs\,dS)(Rs) \;=\; dS\,\big((Rc)(Rc) + (Rs)(Rs)\big).$$

That is, $\mathrm{Im}(d\psi \cdot \overline{\psi}) = dS\,|\psi|^2$.

Remark 3.2. Over a field, dividing by $|\psi|^2 \neq 0$ gives $\mathrm{Im}(d\psi/\psi) = dS$, the guidance law. The division step is *not* part of the formalization; only the cleared identity of Theorem 3.1 is proved.

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Theorem 3.3 (Cleared quantum-potential prefactor; `Bohm.quantumPotential_cleared`)**.** *For all* $\Delta R, R \in \mathbb{Z}$, $(-\Delta R)(2R) = 2\big((-\Delta R)R\big)$.

This is the statement that $-\frac{1}{2}\,\Delta R/R$ and $-\Delta R/(2R)$ agree after clearing the common denominator $2R$.

The proof of Theorem 3.1 uses three auxiliary lemmas, all proved for arbitrary integers:

Lemma 3.4. (1) *(`Bohm.add_neg_self`)* $a + (-a) = 0$.
(2) *(`Bohm.cross`)* $(dR\,s)(Rc) = (dR\,c)(Rs)$*: the two cross terms of the product rule coincide.*
(3) *(`Bohm.phaseTerm`)* $(Rc\,dS)(Rc) = dS\,\big((Rc)(Rc)\big)$*: a phase term factors out* dS.

Proof of Theorem 3.1. Expand the left side. The two cross terms $(dR\,s)(Rc)$ and $-(dR\,c)(Rs)$ cancel by Lemma 3.4(2),(1). The remaining terms $(Rc\,dS)(Rc) + (Rs\,dS)(Rs)$ equal $dS(Rc)(Rc) + dS(Rs)(Rs)$ by Lemma 3.4(3), and distributivity gives the right side. The Lean proof follows exactly this calculation using rewriting with core `Int` lemmas. □

4. Scope

The formalization proves the five integer identities above and nothing more. In particular it does not formalize: the division $\mathrm{Im}(d\psi/\psi) = dS$; any real or complex analysis; the Schrödinger or continuity equations; quantization of circulation or winding; the oscillator eigenvalue; or any numerical scheme.

5. Correspondence between claims and Lean declarations

Claim in this paper	Lean declaration	File
Theorem 3.1	`Bohm.guidance_cleared`	`BohmIdentities.lean`
Theorem 3.3	`Bohm.quantumPotential_cleared`	`BohmIdentities.lean`
Lemma 3.4(1)	`Bohm.add_neg_self`	`BohmIdentities.lean`
Lemma 3.4(2)	`Bohm.cross`	`BohmIdentities.lean`
Lemma 3.4(3)	`Bohm.phaseTerm`	`BohmIdentities.lean`

Each Lean statement is the displayed formula verbatim, with all variables universally quantified over `Int`. The development uses Lean 4 core only (no Mathlib).

Appendix A. Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/bohm-identities`
- Verified commit:
  `6e24d96576d0d54e28ebf882bec19ac56c5a5af6`
- Lean toolchain: `leanprover/lean4:v4.22.0`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/bohm-identities/actions/runs/41794833590`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/bohm-identities/archive/6e24d96576d0d54e28ebf882bec19ac56c5a5af6.tar.gz`
  `21ae9b7cfa87fb56aeca37596df4c0733d6093309540496913081fb16d243437`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 5 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`Bohm.add_neg_self`	`[propext]`
`Bohm.cross`	`[propext]`
`Bohm.guidance_cleared`	`[propext]`
`Bohm.phaseTerm`	`[propext]`
`Bohm.quantumPotential_cleared`	`[propext]`

References

[1] D. Bohm, *A suggested interpretation of the quantum theory in terms of "hidden" variables. I*, Phys. Rev. **85** (1952), 166–179.

[2] E. Madelung, *Quantentheorie in hydrodynamischer Form*, Z. Phys. **40** (1927), 322–326.

[3] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, LNCS 12699 (2021), 625–635.

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